WORKED SOLUTION

STEP I 2015, Question 2

Pure • Trigonometric Solution of Cubics

STEP I 2015, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q2

The key idea

The cubic $4x^3 - 3x$ is exactly what the triple-angle formula produces: $\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha$. So a cubic of this shape can be solved by guessing a cosine. Once we know one root, we can factorise and solve the leftover quadratic. In the final part a rescaling $y = 2x$ turns the given cubic into this shape, and part (i) supplies the exact values we need.

Solution

Part (i)

Write $15^\circ = 45^\circ - 30^\circ$ and use the compound-angle formulae:

$$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}},$$

as required. In the same way,

$$\sin 15^\circ = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}.$$

Rationalising the denominators gives the equivalent forms

$$\cos 15^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}, \qquad \sin 15^\circ = \frac{\sqrt{6}-\sqrt{2}}{4},$$

which are the handier ones in part (iii).

Part (ii)

Write $c = \cos\alpha$ and $s = \sin\alpha$. Expanding $\cos(2\alpha + \alpha)$,

$$\begin{aligned}\cos 3\alpha &= \cos 2\alpha \cos\alpha - \sin 2\alpha \sin\alpha \\ &= (2c^2 - 1)c - 2s^2 c \\ &= 2c^3 - c - 2(1 - c^2)c \\ &= 4c^3 - 3c.\end{aligned}$$

So $4c^3 - 3c - \cos 3\alpha = 0$: $x = \cos\alpha$ is a root.

For the other roots, replace $\cos 3\alpha$ by $4c^3 - 3c$ and factorise:

$$\begin{aligned}4x^3 - 3x - (4c^3 - 3c) &= 4(x^3 - c^3) - 3(x - c) \\ &= (x - c)(4x^2 + 4cx + 4c^2 - 3).\end{aligned}$$

Ask yourself How do we simplify the discriminant of the quadratic factor?

The discriminant is $16c^2 - 16(4c^2 - 3) = 48(1 - c^2) = 48s^2$, using $c^2 + s^2 = 1$. Its square root is $4\sqrt{3}\,|s|$, but since we take both signs $\pm$ we can drop the modulus. The quadratic formula gives

$$x = \frac{-4c \pm 4\sqrt{3}\,s}{8},$$

so the other two roots are

$$x = -\tfrac{1}{2}\cos\alpha + \tfrac{\sqrt{3}}{2}\sin\alpha \quad \text{and} \quad x = -\tfrac{1}{2}\cos\alpha - \tfrac{\sqrt{3}}{2}\sin\alpha.$$

These are $\cos(\alpha - 120^\circ)$ and $\cos(\alpha + 120^\circ)$. That makes sense: $3(\alpha \pm 120^\circ) = 3\alpha \pm 360^\circ$, so these angles have the same $\cos 3\theta$ as α.

Part (iii)

Ask yourself The given cubic is $y^3 - 3y - \sqrt{2}$, not $4x^3 - 3x - \cdots$. How do we match the two?

We want the cubic and linear coefficients to be in the ratio 4 to 3, as in part (ii). Put $y = 2x$:

$$8x^3 - 6x - \sqrt{2} = 0 \iff 4x^3 - 3x - \frac{\sqrt{2}}{2} = 0.$$

This is the equation of part (ii) with $\cos 3\alpha = \frac{\sqrt{2}}{2} = \cos 45^\circ$, so we may take $\alpha = 15^\circ$. Part (ii) then gives three roots x. They are distinct (see below), so they are all the roots of the cubic, and $y = 2x$ gives

$$y = 2\cos\alpha, \qquad y = -\cos\alpha \pm \sqrt{3}\sin\alpha, \qquad \alpha = 15^\circ.$$

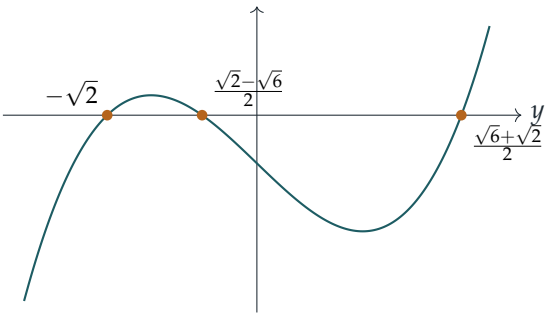
Now substitute the values from part (i).

- $y = 2\cos 15° = \dfrac{\sqrt{3}+1}{\sqrt{2}} = \dfrac{\sqrt{6}+\sqrt{2}}{2}$.
- $y = -\cos 15° + \sqrt{3}\sin 15° = \dfrac{-(\sqrt{6}+\sqrt{2})+(\sqrt{18}-\sqrt{6})}{4}$, which simplifies to $\dfrac{2\sqrt{2}-2\sqrt{6}}{4} = \dfrac{\sqrt{2}-\sqrt{6}}{2}$.
- $y = -\cos 15° - \sqrt{3}\sin 15° = \dfrac{-(\sqrt{6}+\sqrt{2})-(3\sqrt{2}-\sqrt{6})}{4} = -\sqrt{2}$.

Here we used $\sqrt{3}\sqrt{6} = \sqrt{18} = 3\sqrt{2}$. These three values are clearly different. So the solutions are

$$y = \frac{\sqrt{6}+\sqrt{2}}{2}, \qquad y = \frac{\sqrt{2}-\sqrt{6}}{2}, \qquad y = -\sqrt{2}.$$

In terms of angles they are $2\cos 15°$, $2\cos 255°$ and $2\cos 135°$.



Checks

- The y^2 coefficient of $y^3 - 3y - \sqrt{2}$ is zero, so the roots must sum to zero. Indeed $\frac{\sqrt{6}+\sqrt{2}}{2} + \frac{\sqrt{2}-\sqrt{6}}{2} - \sqrt{2} = 0$.
- The product of the roots should be $\sqrt{2}$. Since $(\sqrt{2}+\sqrt{6})(\sqrt{2}-\sqrt{6}) = 2-6 = -4$, the product is $(-1)\cdot(-\sqrt{2}) = \sqrt{2}$.
- The root $y = -\sqrt{2}$ is easy to check directly: $-2\sqrt{2} + 3\sqrt{2} - \sqrt{2} = 0$. Numerically, $\frac{\sqrt{6}+\sqrt{2}}{2} \approx 1.93185$, and $(1.93185)^3 - 3(1.93185) \approx 1.41421 = \sqrt{2}$.
- In part (ii), $\alpha = 0$ gives the equation $4x^3 - 3x - 1 = 0$, i.e. $(x-1)(2x+1)^2 = 0$. The formulae give roots $1, -\frac{1}{2}, -\frac{1}{2}$, as they should.

Common slips

- Forgetting to rescale in part (iii): solving $4x^3 - 3x - \sqrt{2} = 0$ instead. That equation would need $\cos 3\alpha = \sqrt{2} > 1$, which is impossible.
- Giving the answers for x rather than for $y = 2x$.

- Writing $\sqrt{3(1-c^2)} = \sqrt{3}\sin\alpha$ without noting that the $\pm$ sign takes care of the case $\sin\alpha < 0$.

Beyond STEP

This is the trigonometric method for solving a cubic with three real roots, due to Viète. Any cubic can be shifted to remove its square term, giving $t^3 + pt + q = 0$. When $p < 0$, the scaling $t = 2\sqrt{-p/3}\,x$ turns it into $4x^3 - 3x = k$. Cardano's formula fails to help here, because it routes through square roots of negative numbers even though all three roots are real. Seeing why that happens, and why the cosine trick sidesteps it, leads naturally to complex numbers and the roots of unity. Needham's *Visual Complex Analysis* gives a beautiful geometric picture of this.

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